

Six Facts About (Rental) Prices

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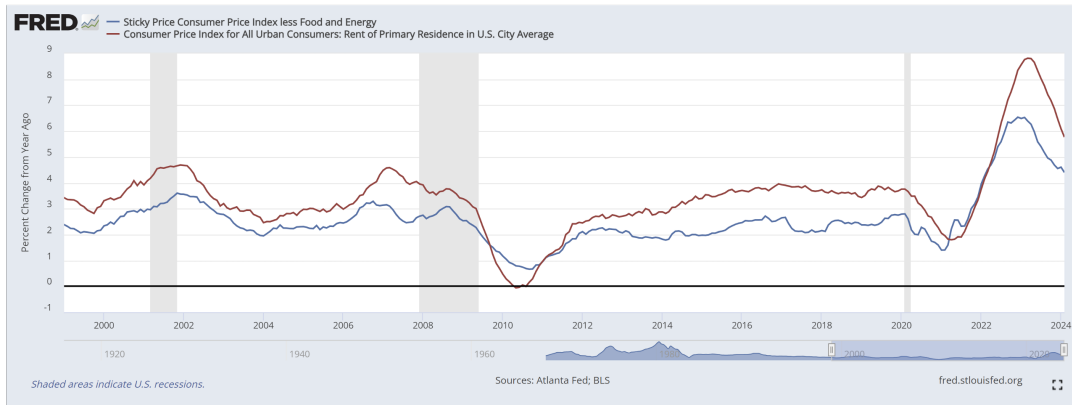
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Rental Prices in the United States

- Rent is largest component of CPI → vital for understanding aggregate prices
- Housing costs are very cyclical → important for dynamics
- Important for understanding **secular changes in economy over time**
 - ▶ What are the welfare implications of entry of large/corporate landlords?
- Little systematic collection of *contract* rent prices → hard to study
 - ▶ Zillow/Craigslist/MLS: **posted** prices, limited time series, and selection of units into sample

Rental Prices in the United States



- Housing costs are the most cyclical part of CPI

What We Are Doing

■ Study rental price setting in Berkeley, California

- ▶ Systematic data collection of unit-level rents over almost 30 years
- ▶ Document novel facts about rental price setting

■ Theoretically:

- ▶ Institutional setting closely mimics Calvo Pricing
- ▶ Rare opportunity to understand firm price setting
 - We know exact form of price adjustment costs!

■ Findings:

- ▶ Significant bunching in rent distribution, price rigidity among landlords free to set new prices
- ▶ Empirics very hard to reconcile with efficient, profit-maximizing behavior by landlords
- ▶ Evidence that large landlords are much more sophisticated than mom-and-pops
- ▶ Imply significant consumer welfare losses from takeover by large landlords

Previous Work

- Prices: Bils and Klenow (2004), Nakamura and Steinsson (2008)
- “Sticky Rents”: Genesove (2003), Aysoy et al. (2014), Hoffmann and Kurz-Kim (2006), Shimizu et al. (2010), Gallin and Verbrugge (2019)
- New Tenant Rent vs. Continuing Tenant Rent: Adams et al. (2024)
- Behavioral Firms: DellaVigna and Gentzkow (2019), Stevens (2020), Reis (2006), Matějka (2016)
- Round-Number Bunching: Dube et al. (2018)

Outline

1 Institutional Details

2 Rental Housing Market in Berkeley

3 Facts About Rent Setting

4 Next Steps

Rent Control in Berkeley, CA

- 26K total rented units in Berkeley, 19K covered by rent control
 - ▶ Rest are partially or fully exempt
- Since Jan. 1999, landlords able to set prices at any rate for new vacancies
- Landlords allowed small adjustment (less than CPI) each year for continuing tenants
- Very strong tenant protections → very hard to evict or force out

Institutional Setting

Calvo Pricing:

- Once tenant in the unit, prices can be raised by small amounts
- Once tenant leaves, landlord completely free to set rent

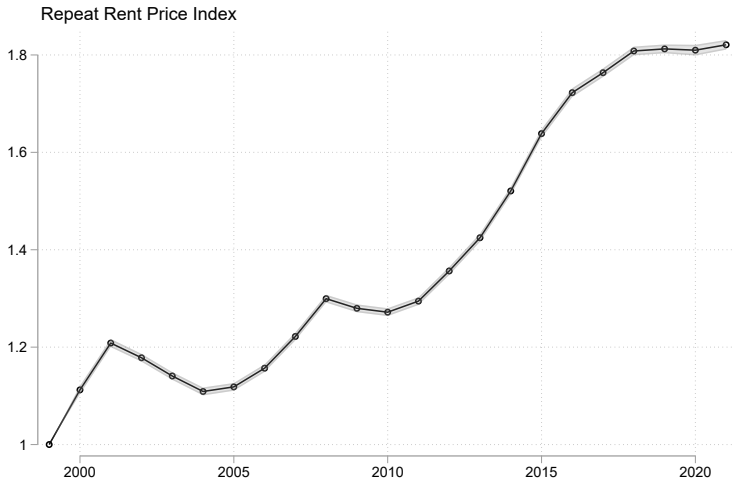
Other Potential Constraints on Price Setting Not Relevant:

- Need to post vacancy either way → no **additional** menu cost
- New tenant → no **existing** implicit contract or long-term relationship

Outline

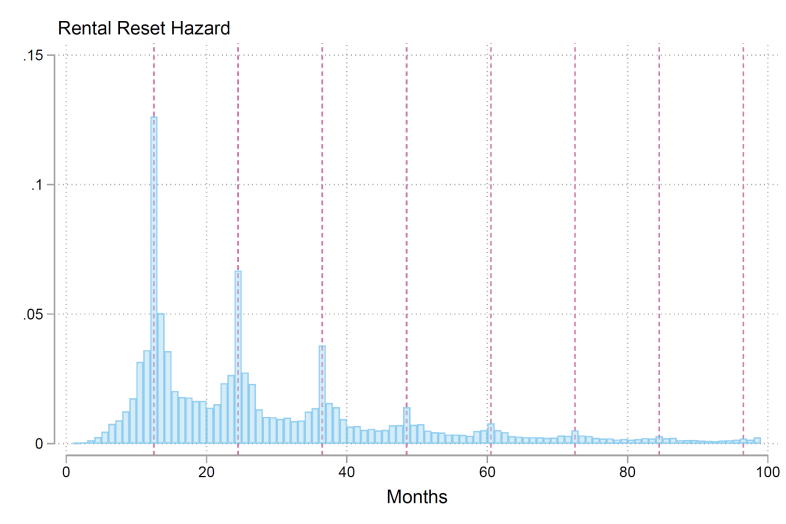
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Strong Growth in Rents for New Tenants



■ Significant periods after recessions where rents decline/flat

Hazard Rate of Rent Price Resets

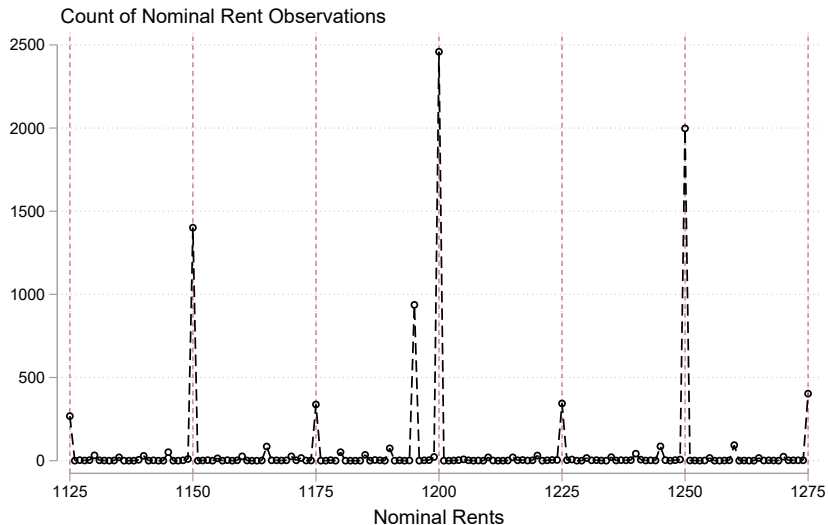


■ Most tenant spells last 1 or 2 years, tenants exit on the year

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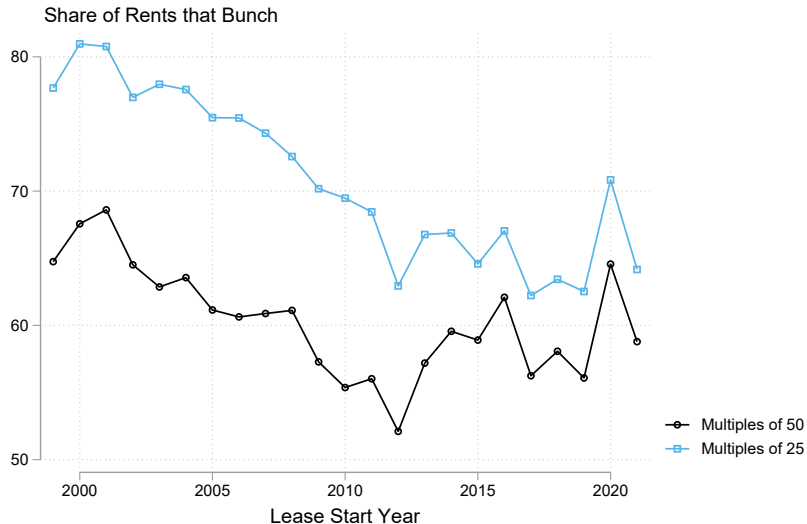
Fact 1: Landlords Bunch Rents at Multiples of 25/50/100



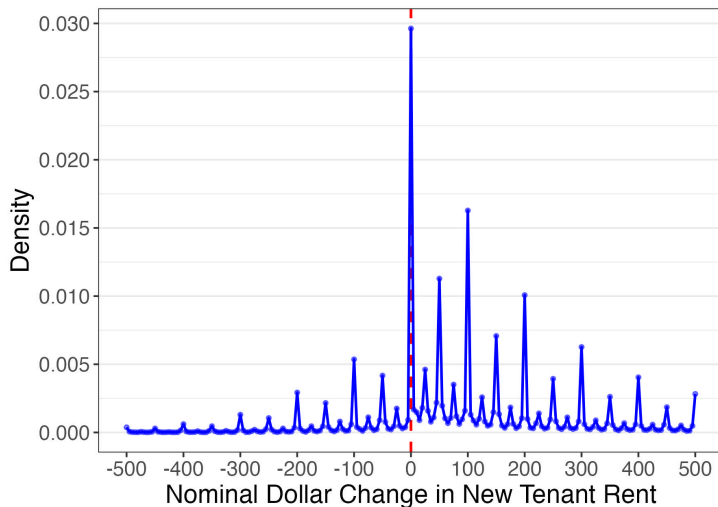
Rent Bunching

- 35% of Rents are multiple of 100, 60% are multiple of 50, 70% are multiple of 25
 - ▶ 97% end in 0 or 5
- Much larger than bunching in wage distribution in admin data Dube et al. (2018)
- Implies significant **market power** and **landlord misoptimization**
 - ▶ Unless tenants strongly prefer round rents

Bunching Decreasing Over Time



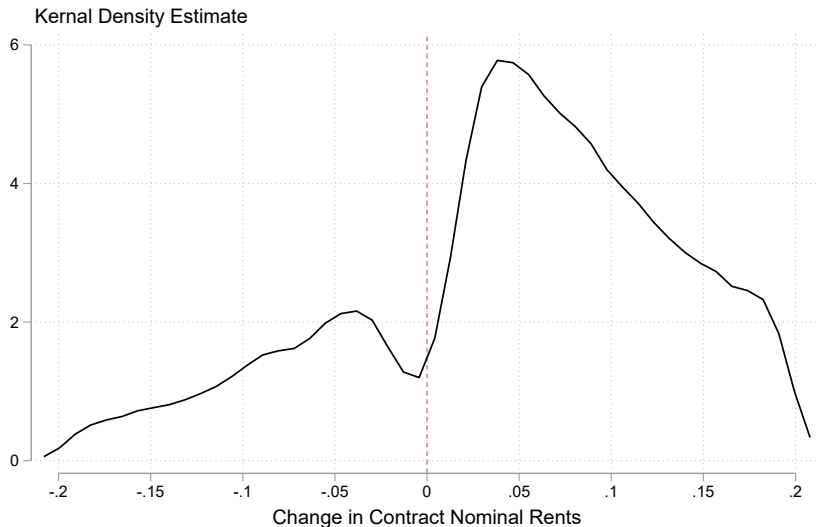
Fact 2: Large, Asymmetric Spike in Rent Change Distribution at Zero



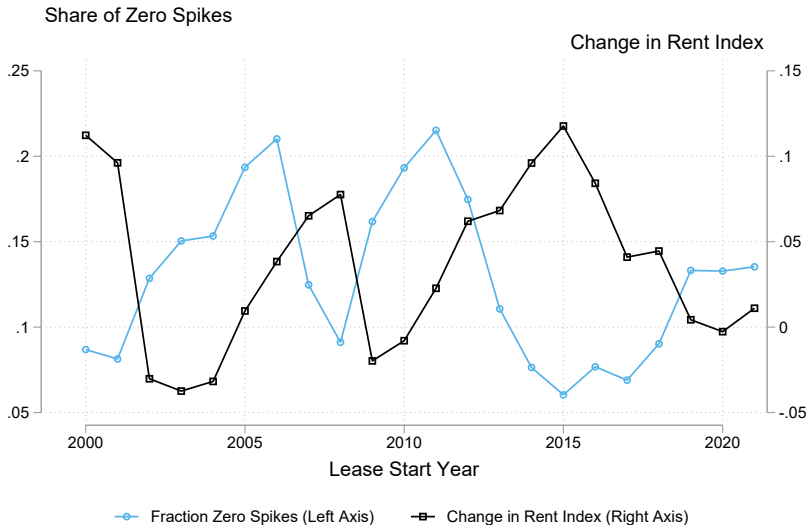
Rent Bunching

- 13% of landlords do not change rents between tenants, despite being free to do so
 - ▶ Potentially very costly choice!
 - ▶ Tenants can stay for years, with very large price growth for **new tenants only**
- “Missing Mass” below zero: 2.5x as many price changes between $(0, 25]$ as $[-25, 0)$

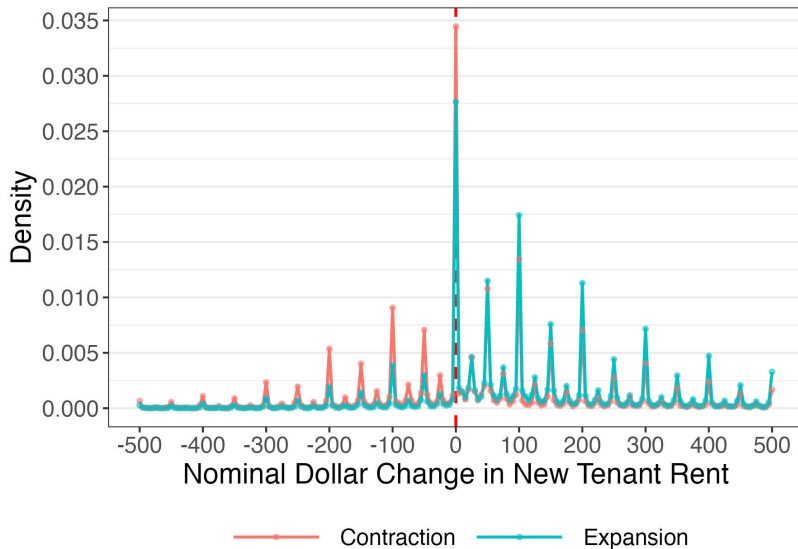
Clear Evidence of Missing Mass in Density (Excluding Zeros)



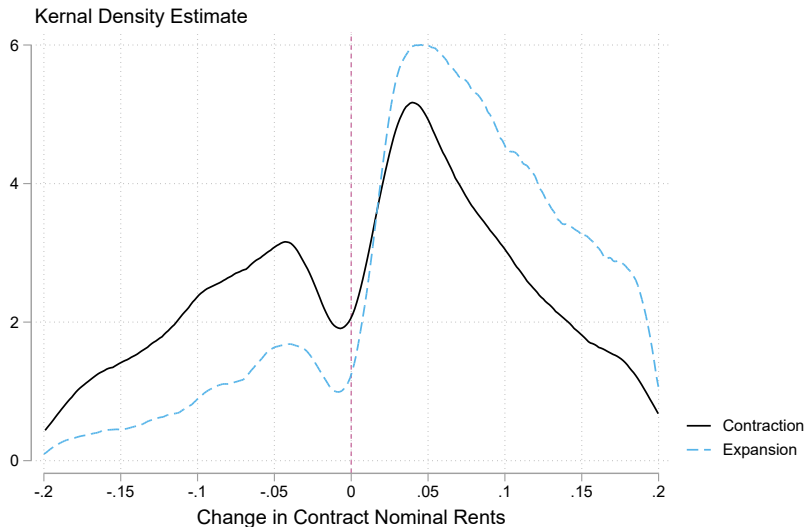
Zero Rent Changes Spike Around Contractions



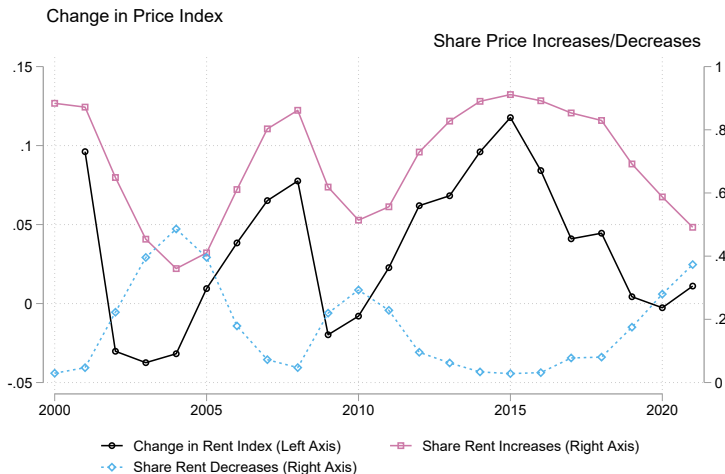
Fact 3: More Price Decreases in Non-Expansions



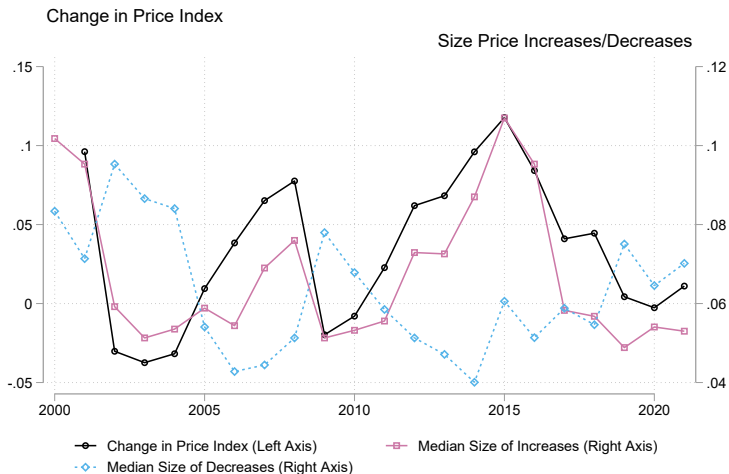
Less Evidence of “Missing Mass” During Contractions (Excluding Zeros)



Fact 4: Share of Price Changes Strongly Correlated With Cycle



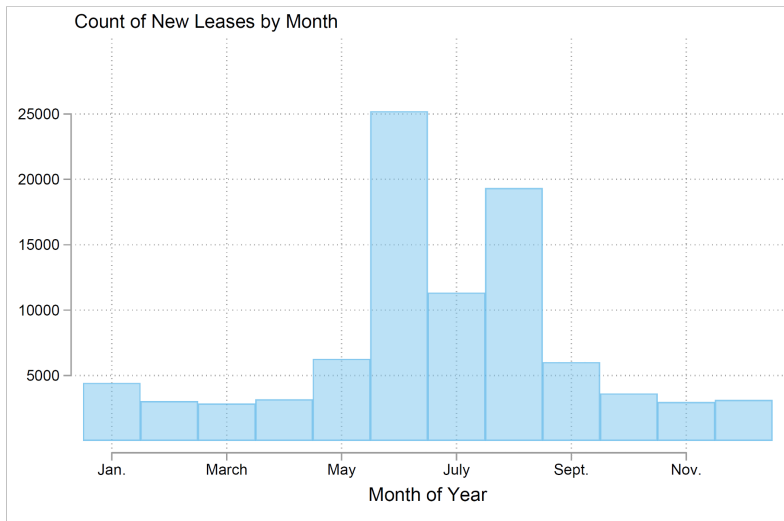
Fact 4: Magnitude of Price Changes Strongly Correlated With Cycle



Size and Frequency of Price Changes

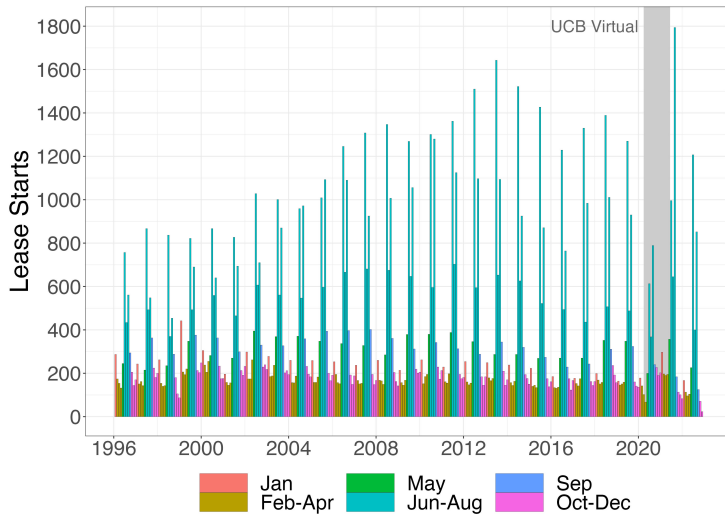
- Size/Share of **both** price increases and decreases strongly correlated with cycle
- Contrasts strongly with the consumer goods market
 - ▶ Only frequency of price increases correlated with cycle (Nakamura & Steinsson, 2008)
 - ▶ Argue consistent with menu-cost model
 - ▶ In contrast, Berkeley rental market is closer to Calvo pricing.

Fact 5: Predictable Seasonal Demand for Rental Units Influences Prices

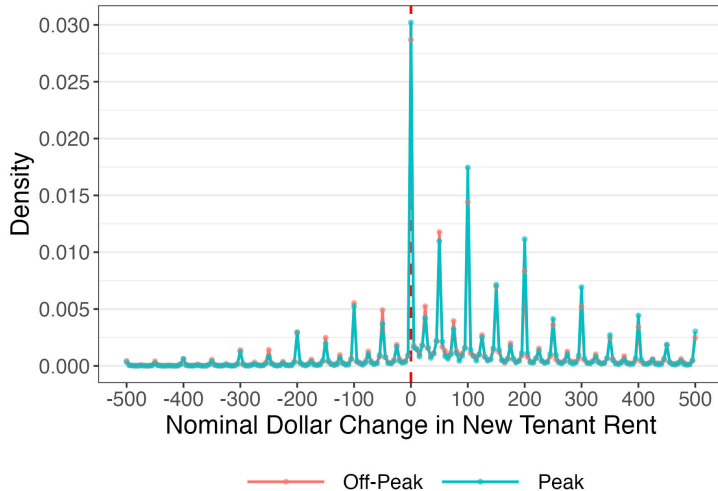


■ Most leases start in summer months

Predictable Seasonal Demand for Rental Units



Price Change Distribution for Peak vs. Off Peak Rents



■ More price decreases, smaller increases

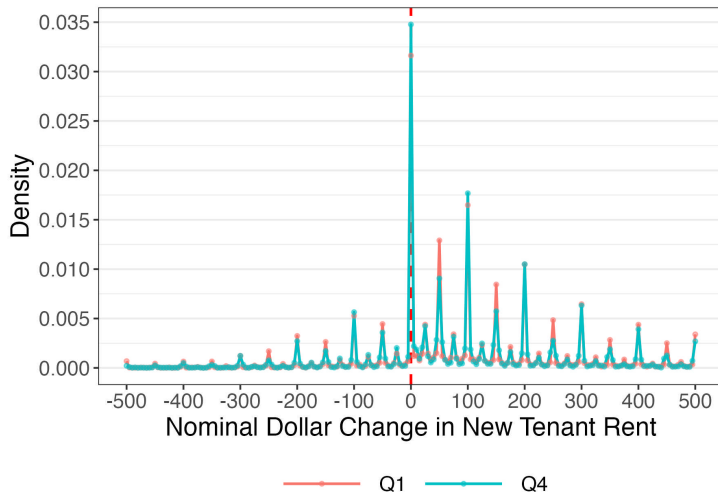
Off Peak Rentals: More Zeros, Price Cuts, and a 4% Rent Discount

	Zero Spike	Price Cut	Big Increase	Log(Rent)
1(Off Peak)	0.0065** (0.0028)	0.038*** (0.0031)	-0.025*** (0.0032)	-0.035*** (0.0013)
<i>N</i>	88724	88724	88724	88724
R^2	0.28	0.32	0.46	0.94
Nbhood-YR FE	Yes	Yes	Yes	Yes
Unit-FE	Yes	Yes	Yes	Yes
Dep. Mean	0.13	0.17	0.28	7.31

Standard errors in parentheses

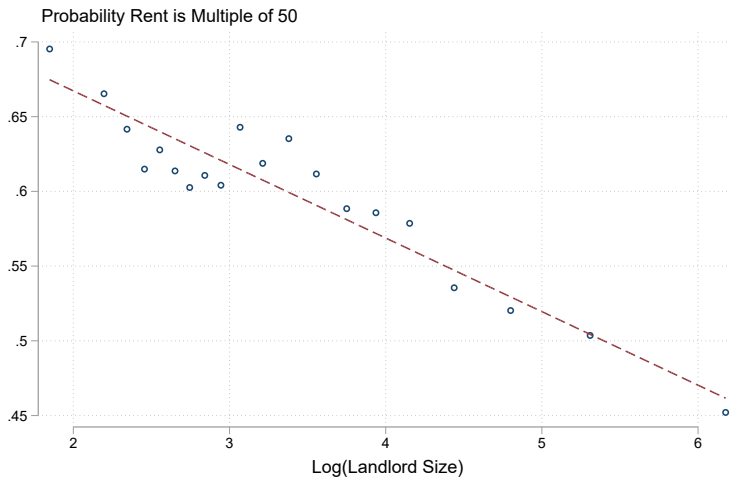
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Fact 6: Heterogeneous Pricing Behavior by Landlord Type



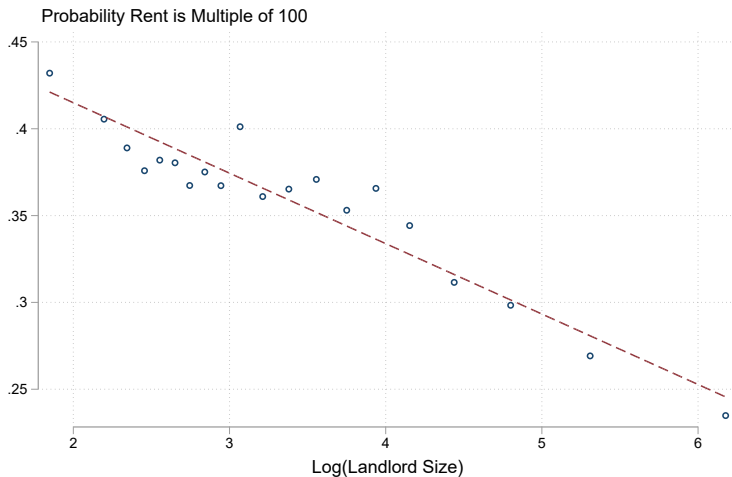
■ Q1 = landlords in smallest size quartile; Q4 = largest

Large Landlords Less Likely to Bunch at Multiples of 50



■ Largest landlords $\implies$ half as much bunching

Large Landlords Less Likely to Bunch at Multiples of 100



■ Largest landlords $\implies$ half as much bunching

Big Landlords Give Fewer Price Cuts and More Large Increases

	Rent Change is...			Rent is...	
	Zero Spike	Price Cut	Big Increase	Multiple 100	Multiple 50
Big Landlord	-0.024*** (0.0038)	-0.011*** (0.0037)	0.016*** (0.0034)	-0.12*** (0.0067)	-0.13*** (0.0069)
<i>N</i>	76958	76958	76958	76958	76958
R^2	0.10	0.32	0.19	0.047	0.054
Nbhood-YR FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.14	0.26	0.18	0.35	0.58

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Heterogeneous Pricing Over Cycle

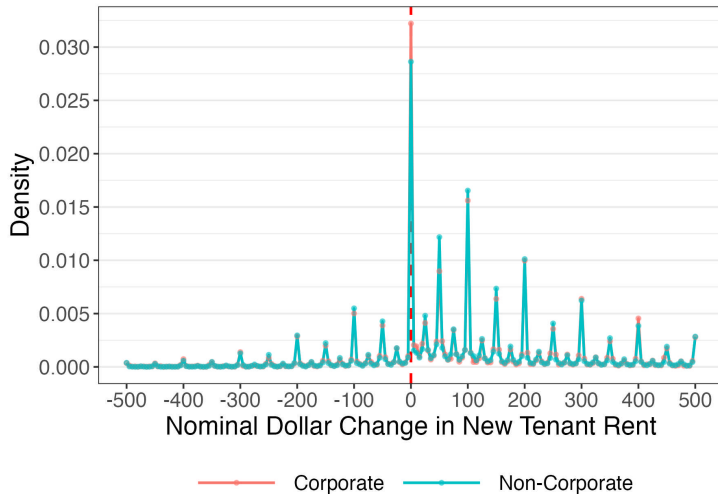
	Rent Change is...			Rent is...	
	Zero Spike	Price Cut	Big Increase	Multiple 100	Multiple 50
Big Landlord	-0.030*** (0.0053)	-0.018*** (0.0041)	0.037*** (0.0056)	-0.12*** (0.0078)	-0.13*** (0.0080)
Big Landlord x Cycle	0.016** (0.0070)	0.022*** (0.0080)	-0.056*** (0.0077)	-0.014 (0.011)	-0.0021 (0.011)
<i>N</i>	75763	75763	75763	75763	75763
<i>R</i> ²	0.100	0.32	0.19	0.046	0.052
Nbhood-YR FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.14	0.26	0.18	0.35	0.59

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

■ Coef. on cycle: effect when rent price index increases by 10%

Pricing Differences by Corporate Status



■ Corporate defined as names with “LLC,” “Inc,” , “Corp”

Corporate Landlords Give More Large Increases

Table: Rent-Setting by Corporate Landlord Status

	Rent Change is...			Rent is...	
	Zero Spike	Price Cut	Big Increase	Multiple 100	Multiple 50
Corporate	-0.005* (0.003)	0.002 (0.003)	0.017*** (0.003)	-0.082*** (0.005)	-0.004 (0.004)
Nbhd-Yr FE	Y	Y	Y	Y	Y
Dep. Mean	0.13	0.17	0.28	0.23	0.35
Observations	97,018	97,018	97,018	97,018	97,018
Adjusted R ²	0.090	0.172	0.325	0.040	0.016

Note:

*p<0.1; **p<0.05; ***p<0.01

■ Corporate defined as names with “LLC,” “Inc,” , “Corp”

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Model & Extensions (In Progress)

- Follow Dube et al. (2018): left-digit renter bias or behavioral firms?
- Experiment with Berkeley students: preference for round number rents?
- Provide evidence of extent of monopoly power & mispricing
- Add behavioral parameters to calvo model
- Quantify consumer welfare loss of movement to large landlords

Conclusion I

1 Strong evidence of heuristic pricing

- ▶ Rents bunch at multiples of 25/50/100
- ▶ Evidence of very strong monopoly power & landlord misoptimization

2 Evidence of nominal rent rigidity for new tenants, even without any rent control

- ▶ Large, asymmetric spike in rent change distribution at zero
- ▶ Firms don't change prices, even when able and incentives to do so are high

3 Landlords accept rent cuts in contractions

- ▶ Observe more price decreases in non-expansions
- ▶ Rent rigidity less binding than zero spike suggests

Conclusion II

- 4 Frequency and magnitude of both price increases & decreases related to the business cycle (unlike for retail goods)
- 5 Seasonal demand has large effects on magnitude and variance of price changes
- 6 Larger/corporate landlords use fewer heuristics → more sophisticated
 - ▶ Implies substantial consumer welfare changes if landlord composition changes
 - ▶ Also depends on whether profitability for large landlords increases rental housing supply
 - ▶ Landlord size also predictive of responsiveness to the business cycle

References I

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